

4	(a) total resistance = $20 \text{ (k}\Omega\text{)}$ current = $12 / 20 \text{ (mA)}$ or potential divider formula p.d. = $[12 / 20] \times 12 = 7.2 \text{ V}$	C1 C1 A1	[3]
	(b) parallel resistance = $3 \text{ (k}\Omega\text{)}$ total resistance $8 + 3 = 11 \text{ (k}\Omega\text{)}$ current = $12 / 11 \times 10^3 = 1.09 \times 10^{-3}$ or $1.1 \times 10^{-3} \text{ A}$	C1 C1 A1	[3]
	(c) (i) LDR resistance decreases total resistance (of circuit) is less hence current increases	M1 A1	[2]
	(ii) resistance across XY is less less proportion of 12 V across XY hence p.d. is less	M1 A1	[2]
5	(a) $E \equiv \text{stress} / \text{strain}$	B1	[1]